

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase.

INPUT	PROCESS	OUTPUT
• num_notebooks • num_pens • payment	$\text{notebookCost} \leftarrow \text{num_notebooks} \times 45$ $\text{penCost} \leftarrow \text{num_pens} \times 12$ $\text{totalCost} \leftarrow \text{notebookCost} + \text{penCost}$ $\text{change} \leftarrow \text{payment} - \text{totalCost}$	• totalCost - change

DECOMPOSITION:

1. Read the number of notebooks, number of pens, and payment amount.
2. Calculate the notebook subtotal ($\text{num_notebooks} \times 45$).
3. Calculate the pen subtotal ($\text{num_pens} \times 12$).
4. Calculate the total cost by adding the notebook and pen subtotals.
5. Calculate the change by subtracting the total cost from the payment.
6. Display the total cost and change.

PSEUDOCODE:

```
START
READ num_notebooks, num_pens, payment
notebookCost  $\leftarrow$  num_notebooks  $\times$  45
penCost  $\leftarrow$  num_pens  $\times$  12
totalCost  $\leftarrow$  notebookCost + penCost
change  $\leftarrow$  payment - totalCost
DISPLAY totalCost, change
END
```

DRY-RUN CALCULATION: QUICK ACTIVITY ONE

CASE TEST ONE:

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays ₱200.

STEP	VARIABLE	CALCULATION	RESULT
1	notebookCost	$3 \times ₱45$	₱135.00
2	penCost	$2 \times ₱12$	₱24.00
3	totalCost	$₱135 + ₱24$	₱159.00
4	change	$₱200 - ₱159$	₱41.00

EXPECTED OUTPUT:

Total cost: ₱159.00

Change: ₱41.00

CASE TEST TWO:

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays ₱150.

STEP	VARIABLE	CALCULATION	RESULT
1	notebookCost	$2 \times ₱45$	₱90.00
2	penCost	$3 \times ₱12$	₱36.00
3	change	$₱90 + ₱36$	₱126.00
4	change	$₱150 - ₱126$	₱24.00

EXPECTED OUTPUT:

Total cost: ₱126.00

Change: ₱24.00